ADIB HASSAN

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Education

- **BSc in Electrical & Electronic Engineering** — *Rajshahi, Bangladesh*
  Rajshahi University of Engineering & Technology (RUET) — *2022– 2026*
  CGPA: 3.79 out of 4.00
  Average CGPA of Last 4 Semesters: 3.85 out of 4.00
  Class Rank: 15th out of 181 students
- **Higher Secondary Certificate (HSC)** — *Dhaka, Bangladesh*
  NotreDame College — *2018– 2020*
  Group: Science
  GPA: 5.00 out of 5.00

Research Interest

- Photonic Crystal Fiber (PCF)
- Nanophotonics
- Plasmonic and Photonic Biosensing
- Wireless Communications
- Biomedical Device
- Metamaterials

Research Experience

Undergraduate Researcher

Undergraduate Research Work, Dept. of EEE, RUET — *November 2024–Present*

- Designed and characterized dual-core, dual-channel photonic crystal fiber (PCF) surface plasmon resonance (SPR) biosensors for label-free, refractive-index-based biomedical sensing, targeting cancer, dengue, tuberculosis, urinary tract infection (UTI), and multi-analyte disease detection.
- Designed and optimized multilayer SPR biosensors incorporating 2D nanomaterials and material interfaces with emphasis on improving sensitivity, detection range, figure of merit, and sensing performance.
- Contributed to secure outage performance & physical layer security analysis of RIS-assisted wireless networks, incorporating SC diversity and MVDR beamforming over κ-μ shadowed fading channels, with analysis of SOP, IP, and PNZSC
- Authored full manuscripts in IEEE & Elsevier style for journal and conference submission.
- **Thesis Title:** "Highly Sensitive N-FK51A/Cu/ZnSe/BP-Based SPR Biosensor for Dengue Virus Detection: A TMM and FEM Approach"
  Supervisor: Tasnim Sarker Joyeeta, Assistant Professor, Dept. of EEE, RUET

Publications

- **A. Hassan**, A. H. Moon, and T. Ahmed, "Highly Sensitive N-FK51A/Cu/ZnSe/BP-Based SPR Biosensor for Dengue Virus Detection: A TMM and FEM Approach," in 2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE), 2026: IEEE, pp. 1-6.
- M. A. H. Moon, **A. Hassan**, and T. Ahmed, "Design and optimization of a CaF_2/Cu/GO/InN multilayer SPR biosensor for label-free and highly sensitive tuberculosis diagnosis," in 2025 IEEE International Conference on Telecommunications and Photonics (ICTP), Dec. 2025, pp. 1-5.
- M. A. Chowdhury, M. S. Hossain, **A. Hassan**, and A. Habib, "Physical Layer Security Analysis of RIS-Assisted Multicasting with MVDR Beamforming Over κ-μ Shadowed Fading Channels," in Proc. 2026 IEEE International Conference on Microwave, Antennas, RF and Sensors (MARSCON), Dhaka, Bangladesh, Aug. 13–14, 2026.
- S. K. Safit, M. Z. I. Sarkar, P. Mahmud, A. Habib, and **A. Hassan**, "Secure outage performance analysis of reconfigurable intelligent surface-aided networks with selection combining diversity scheme," in *2026 Int. Conf. Power, Electron., Commun., Comput., Intell. Infrastruct. (PECCII)*, Pabna, Bangladesh, 2026, pp. 1-6.

Publications (Under Review)

- **A. Hassan**, M. T. Hasan, S. Anik, T. Ahmed, and A. K. Paul, "High-performance dual-channel PCF-SPR biosensor with broad RI detection range for multi-category disease detection," *Optik*, Manuscript under review.
- **A. Hassan,** T. S. Joyeeta, and S. Anik, "High-performance dual-channel PCF-SPR biosensor for broad-range multi-analyte biomedical sensing," submitted to the *14th International Conference on Electrical and Computer Engineering (ICECE 2026)*, Dhaka, Bangladesh, 2026. Manuscript under review**.**

Professional Experience

March 2, 2025– March 9, 2025

Industrial Attachment

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Siliconova Limited (Semiconductor & VLSI Technology Company)

• Worked with Verilog RTL coding and simulation, ASIC physical design, Design for Testability (DFT), timing analysis, synthesis flow, and Cadence IC design tools.

Projects

- Autosentinel: Arduino-powered ultrasonic radar with precision targeting and firing system.
- PyroGuard: An intelligent fire-fighting robot for Rapid Emergency Response.
- Smart Arduino-Powered Parking Management & Monitoring System.
- Line Follower and Obstacle Avoidance Robot using Arduino Uno and IR Sensors
- Electrical Service Design using AutoCAD: a complete electrical service design for a modern apartment building.

Extra-curricular Activities & Achievements

- **IEEE RUET STUDENT BRANCH** ***2022-2026***
  Chair **[PDF]**
  Served as the Chair of IEEE RUET SB, overseeing team activities and orchestrating major events (Etechnovation, Technofair) and workshops as part of the organizing committee, including several conferences (ICCIT, ICECTE, PEEIACON, INDcon), and a national event named RUET EEE Day.
  - Founding Member of **IEEE RUET AP-S** Student Branch Chapter.
  - Former **Treasurer** at IEEE RUET SB
  - Former **Publicity Coordinator** at IEEE RUET SB
- **IEEE Computer Society Bangladesh Chapter** ***2026-Present***
  Vice Chair (Activity)
- **HULT PRIZE at RUET** ***2023-2025***
  Director (Event Management)
- **RUET CORE (Collaborative Opportunities in Research & Education)** ***2024-2026***
  Vice Chair (Outreach) & Founding Member **[PDF]**
- **NDCAR (NotreDame College Association of RUET)** ***2024-2026***
  Joint Secretary
- **Technical Scholarship** (2023-present) from Rajshahi University of Engineering and Technology (RUET), Rajshahi, Bangladesh.
- **Board Scholarship** of Junior School Certificate (JSC) Examination, General Grade.
- "**Perfect Attendance Certificate**" from NotreDame College.

Technical Skills

- **Engineering Software:** COMSOL Multiphysics, LOGO! Soft Comfort, Microwind, Multisim, Arduino IDE, Omron CX-Programmer, Anaconda Navigator.
- **Programming:** MATLAB, C, Python.
- **Microsoft Office:** Microsoft Word, Microsoft Excel, Microsoft PowerPoint.

References

Dr. Tanvir Ahmed
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Dr. Md. Selim Hossain
Professor,
Department of Electrical & Electronic Engineering, RUET
Email: prof.selim@eee.ruet.ac.bd